Rudivan Paixão Barbosa

Decision Science Analyst · Cassotis Consulting
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SUMMARY

Professional trained and working in optimization and operations research, with experience in mathematical modeling (linear, mixed-integer, nonlinear and mixed-integer nonlinear programming) and in implementing algorithms in C/C++. Experience with optimization tools such as Concert Technology, AMPL, CPLEX and Hexaly, and with classical metaheuristics (GRASP, Genetic Algorithms, VNS, ILS, LAHC, Tabu Search), including parameter tuning with iRace. Complementary training in asymptotic analysis of algorithms and in multiobjective optimization techniques (Weighted Sum Method and ε-constraint Method). Applies these skills professionally to the development of mathematical models for the steel industry. Research on lot-sizing and scheduling, vehicle routing and facility location problems.

PROFESSIONAL EXPERIENCE

Decision Science Analyst *04/2024 – Present*
Cassotis Consulting · Belo Horizonte, MG, Brazil

- Development, support and improvement of mathematical programming models for the largest steel companies in Latin America.
- Skills applied: mixed-integer optimization, nonlinear optimization, mathematical modeling, Git, AMPL and LocalSolver.

Substitute Professor *09/2020 – 10/2021*
Federal University of Espírito Santo (UFES), Department of Applied Mathematics · São Mateus, ES, Brazil

- Courses taught: Differential and Integral Calculus, Linear Algebra, Analytic Geometry and Basic Mathematics.

EDUCATION

PhD in Computer Science (Algorithms and Mathematical Optimization) *03/2022 – Present*
Federal University of Ouro Preto (UFOP)

- Thesis: Location-Allocation and Routing Problem for Mammography Units.
- Advisor: Marcone Jamilson Freitas Souza.

Master's in Computational Mathematics (Mathematical Optimization) *08/2017 – 08/2019*
University of São Paulo (USP)

- Dissertation: Solution approaches for the lot-sizing and scheduling problem with order acceptance.
- Advisor: Maristela Oliveira dos Santos.

Specialization in Mathematics Teaching *01/2024 – 06/2024*
UniBF College

Teaching Degree in Mathematics *01/2024 – 09/2024*
UniBF College

Bachelor's in Mathematics (Operations Research and Optimization) *03/2012 – 01/2017*
Federal University of Espírito Santo (UFES)

- Undergraduate thesis: Optimal design of a semi-mechanized network.
- Advisor: Gilberto de Miranda Junior.

AREAS OF EXPERTISE AND SKILLS

Area/Subarea: Computer Science, Optimization.

Research topics: lot-sizing and scheduling, vehicle routing, facility location, metaheuristics.

Tools and languages: AMPL, CPLEX, Concert Technology, Hexaly, C/C++ and Git.

Methods: linear, mixed-integer, nonlinear and mixed-integer nonlinear programming; GRASP, Genetic Algorithms, VNS, ILS, LAHC, Tabu Search; parameter tuning with iRace; asymptotic analysis of algorithms; multiobjective optimization (Weighted Sum, ε-constraint).

Teaching: Differential and Integral Calculus, Linear Algebra, Analytic Geometry, Basic Mathematics.

LANGUAGES

Portuguese: native.

English: professional working proficiency.

SELECTED PUBLICATIONS

- BARBOSA, R. P. et al. Mammography Unit Location: Reconciling Maximum Coverage and Budgetary Constraints. In: 25th International Conference on Enterprise Information Systems, 2023. Link
- BARBOSA, R. P. et al. Um novo modelo para o problema de dimensionamento e sequenciamento de lotes com aceitação de pedidos [A new model for the lot-sizing and scheduling problem with order acceptance]. In: LV Brazilian Symposium on Operations Research (SBPO), 2023. In Portuguese. DOI
- BARBOSA, R. P. et al. Um modelo para o problema de dimensionamento e sequenciamento de lotes com aceitação de pedidos [A model for the lot-sizing and scheduling problem with order acceptance]. In: LI Brazilian Symposium on Operations Research (SBPO), 2019. In Portuguese. DOI
- RIBEIRO, L. T.M. et al. Dimensionamento e sequenciamento de lotes da produção: insights de uma revisão sistemática da literatura e análise bibliométrica [Production lot-sizing and scheduling: insights from a systematic literature review and bibliometric analysis]. In: ENEGEP 2024, Brazilian National Meeting of Production Engineering, 2024. In Portuguese. DOI